

The Weil Conjectures as Proof of Concept for RH: Frobenius, Purity, and What Is Missing over $\mathbb{Q}$

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Abstract

The Weil conjectures for varieties over finite fields include a Riemann hypothesis: the eigenvalues of Frobenius on ℓ -adic cohomology are algebraic integers of absolute value $q^{w/2}$ (purity). Deligne proved this in 1974 using the geometry of the situation — Hard Lefschetz, the Ramanujan conjecture for forms of weight ≥ 2 , and a global-to-local principle enabled by Frobenius acting as a well-defined, geometric operator. The Riemann hypothesis for $\zeta(s)$ is the $q \rightarrow \infty$ limit of this picture, but that limit is ill-posed: there is no Frobenius over $\mathbb{Q}$, no ℓ -adic cohomology of $\text{Spec } \mathbb{Q}$ with a geometric action, and no purity theorem in the number field setting. The Hilbert-Pólya conjecture proposes the substitute: a self-adjoint operator H whose spectrum is the imaginary parts of the Riemann zeros, playing the role of Frobenius. We make precise what Frobenius does in the function field setting and why its absence over $\mathbb{Q}$ is the structural obstacle.

1 The Weil Zeta Function

Let X be a smooth projective variety over $\mathbb{F}_q$. The *Weil zeta function* is

$$Z(X, T) = \exp \left(\sum_{n=1}^{\infty} \frac{|X(\mathbb{F}_{q^n})|}{n} T^n \right), \quad T = q^{-s}.$$

Theorem 1 (Weil conjectures [1]). $Z(X, T)$ satisfies:

1. **Rationality:** $Z(X, T) \in \mathbb{Q}(T)$.
2. **Functional equation:** $Z(X, 1/q^{d/2}T) = \pm q^{d\chi/2} T^\chi Z(X, T)$, where $d = \dim X$ and $\chi = \chi(X)$ is the Euler characteristic.
3. **Riemann hypothesis:** $Z(X, T) = \prod_{i=0}^{2d} P_i(T)^{(-1)^{i+1}}$ where $P_i(T) = \prod_j (1 - \alpha_{i,j}T)$ with $|\alpha_{i,j}| = q^{i/2}$.

Parts (1) and (2): proved by Dwork (1960) [2] and Grothendieck (1964–1965) [3]. Part (3): proved by Deligne (1974) [4].

2 Frobenius and Purity

The key actor is the Frobenius endomorphism.

Definition 2 (Geometric Frobenius). The *geometric Frobenius* $\text{Frob}_q : X \rightarrow X$ is the map $(x_0 : \cdots : x_n) \mapsto (x_0^q : \cdots : x_n^q)$ on projective coordinates. It is a morphism of $\mathbb{F}_q$ -schemes, fixing $X(\mathbb{F}_q)$ setwise and acting on $X(\mathbb{F}_{q^n})$ with the n -th power counting fixed points: $|X(\mathbb{F}_{q^n})| = |\text{Fix}(\text{Frob}_q^n)|$.

The Lefschetz trace formula (Grothendieck, via ℓ -adic cohomology) gives:

$$|X(\mathbb{F}_{q^n})| = \sum_{i=0}^{2d} (-1)^i \text{Tr} \left(\text{Frob}_q^n \middle| H_{\text{ét}}^i(X_{\overline{\mathbb{F}}_q}, \mathbb{Q}_\ell) \right).$$

Therefore $Z(X, T) = \prod_i \det(1 - T \cdot \text{Frob}_q | H_{\text{ét}}^i)^{(-1)^{i+1}}$, and the $\alpha_{i,j}$ are the eigenvalues of Frobenius on $H_{\text{ét}}^i$.

Theorem 3 (Deligne: Purity [4]). *Let $X/\mathbb{F}_q$ be smooth projective of dimension d . Then for each i , every eigenvalue $\alpha_{i,j}$ of Frob_q on $H_{\text{ét}}^i(X_{\overline{\mathbb{F}}_q}, \mathbb{Q}_\ell)$ is an algebraic integer with $|\alpha_{i,j}| = q^{i/2}$ (for all embeddings into $\mathbb{C}$). This is the Riemann hypothesis for $Z(X, T)$: the zeros and poles of $Z(X, T)$ lie on the circles $|T| = q^{-i/2}$, $i = 0, \dots, 2d$.*

3 How Deligne's Proof Works

The proof of Theorem 3 proceeds in three main steps.

Step 1: Hard Lefschetz. For a smooth projective X of dimension d , cupping with the hyperplane class $\eta \in H^2$ gives an isomorphism $H^i \xrightarrow{\sim} H^{2d-i}$ (Hard Lefschetz, proved by Deligne [5] as part of the Weil II package). This gives the functional equation and controls the Hodge-type structure of the cohomology.

Step 2: Global Ramanujan. Deligne's key input is the Ramanujan conjecture for holomorphic cusp forms of weight $k \geq 2$: $|\tau(p)| \leq 2p^{(k-1)/2}$ (proved in [4] as a consequence of purity for modular curves). The cusp form $\Delta(z) = \sum \tau(n)q^n$ corresponds to the motive h^1 of a specific variety; purity for that variety gives the Ramanujan bound, which in turn propagates to give purity for general X via a global-to-local reduction.

Step 3: Purity propagation. Using Lefschetz pencils (a family of hyperplane sections X_t parametrized by $t \in \mathbb{P}^1$), Deligne reduces the higher-dimensional case to the curve case via a monodromy argument. The Frobenius acts on the cohomology of the fibers; purity for curves (proved first) propagates through the Leray spectral sequence.

Remark 4 (What makes the proof work). Every step of the proof uses Frobenius in an essential way:

- Hard Lefschetz: uses the Frobenius-equivariant Künneth decomposition.
- Ramanujan: the Hecke operators T_p are the arithmetic Frobenii; their eigenvalues are the $\alpha_{i,p}$.

- Purity propagation: the monodromy group is a subgroup of $\mathrm{GL}(H^i)$ normalized by Frobenius; the eigenvalue bound comes from Frobenius acting on the nearby cycles sheaf.

Frobenius is available because the base field is $\mathbb{F}_q$, which has a canonical absolute Galois group $\hat{\mathbb{Z}}$ generated by Frob_q .

4 The Riemann Zeta Function as a Weil Zeta Function

The Riemann zeta function $\zeta(s) = Z(\mathrm{Spec} \mathbb{Z}, q^{-s})$ is *formally* the Weil zeta function of $\mathrm{Spec} \mathbb{Z}$ (the ring of integers of $\mathbb{Q}$). The Weil conjectures predict:

$$\zeta(s) = \frac{P_1(q^{-s})}{P_0(q^{-s}) \cdot P_2(q^{-s})}$$

with P_i having roots of absolute value $q^{i/2}$, and the RH being that the roots of P_1 have $|\alpha| = q^{1/2}$, i.e., $\mathrm{Re}(s) = 1/2$.

The problem: $\mathrm{Spec} \mathbb{Z}$ is not a variety over any $\mathbb{F}_q$. It is a scheme over $\mathrm{Spec} \mathbb{Z}$ itself.

Proposition 5 (No Frobenius over $\mathbb{Q}$). *There is no global Frobenius endomorphism $\mathrm{Frob} : \mathrm{Spec} \mathbb{Z} \rightarrow \mathrm{Spec} \mathbb{Z}$ playing the role of $x \mapsto x^q$ over $\mathbb{F}_q$. Specifically:*

1. Over $\mathbb{F}_q$: Frob_q is a morphism of the ambient scheme, acting on cohomology globally and equivariantly with respect to the Galois group $\mathrm{Gal}(\bar{\mathbb{F}}_q/\mathbb{F}_q) \cong \hat{\mathbb{Z}}$.
2. Over $\mathbb{Q}$: the Frobenius at each prime p is a local element $\mathrm{Frob}_p \in \mathrm{Gal}(\bar{\mathbb{Q}}_p/\mathbb{Q}_p)$, defined only up to conjugacy in $\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$, and there is no canonical global element.
3. The Galois group $\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ is not generated by any single element (unlike $\hat{\mathbb{Z}}$); it is a profinite group of enormous complexity (the absolute Galois group of $\mathbb{Q}$, far from abelian).

Remark 6. Over $\mathbb{F}_q(C)$ (function field of a curve C), the Frobenius at each closed point is well-defined and the global Frobenius Frob_q acts geometrically on C . The point counts $|C(\mathbb{F}_{q^n})|$ are finite and give a polynomial zeta function. For $\mathrm{Spec} \mathbb{Z}$: the “point counts” are $|\mathbb{P}^0(\mathbb{F}_p)| = 1$ for each prime p — trivial. There is no arithmetic geometry making $\mathrm{Spec} \mathbb{Z}$ look like a curve over any $\mathbb{F}_q$, because $\mathbb{Z}$ has characteristic zero.

5 The Hilbert-Pólya Operator as Frobenius Substitute

Definition 7 (Hilbert-Pólya conjecture, geometric form). The Hilbert-Pólya conjecture asserts the existence of a self-adjoint operator H on a Hilbert space $\mathcal{H}$ such that:

$$\zeta\left(\frac{1}{2} + i\lambda\right) = 0 \iff \lambda \in \mathrm{Spec}(H).$$

Proposition 8 (HP as Frobenius substitute). *The Hilbert-Pólya operator H , if it exists, plays the role of Frobenius in the following precise sense:*

- *Frobenius (over $\mathbb{F}_q$): the eigenvalues on $H_{\text{ét}}^1$ are the $\alpha_{1,j}$ with $|\alpha_{1,j}| = q^{1/2}$. The Weil RH says $\alpha_{1,j} = q^{1/2}e^{i\theta_j}$ for real θ_j .*
- *HP (over $\mathbb{Q}$): the zeros of $\zeta(s)$ on $\text{Re}(s) = 1/2$ are $1/2 + i\lambda_j$ for real $\lambda_j \in \text{Spec}(H)$.*

In both cases, the “RH statement” is that the relevant object (Frobenius eigenvalues, or H -eigenvalues) is real (after removing the $q^{1/2}$ or $1/2$ shift). Self-adjointness of H plays the role of purity of Frobenius.

Remark 9 (Evidence from GUE). Montgomery’s pair correlation conjecture [6] predicts that the normalized spacings $\{(\lambda_{n+1} - \lambda_n) \log \lambda_n / 2\pi\}$ follow the GUE distribution — the eigenvalue spacing of a large random Hermitian (self-adjoint) matrix. Odlyzko’s numerics [7] confirm this to high precision. The GUE is the natural ensemble of eigenvalues of a self-adjoint operator, strongly suggesting H exists and is of Wigner-Dyson type.

Remark 10 (Berry-Keating). Berry-Keating [8] propose $H = xp + px$ (quantization of the classical $H_{\text{cl}} = xp$) on an appropriate half-line. The classical orbits of xp are hyperbolic and give the correct leading-order zero density via the Gutzwiller trace formula. The boundary conditions that force the spectrum to match the Riemann zeros exactly are not known. This is the closest concrete proposal for H .

6 Why the Function Field Proof Cannot Transfer

Theorem 11 (Structural obstruction). *Deligne’s proof of purity (Theorem 3) uses three tools that are unavailable for $\zeta(s)$:*

1. **Étale cohomology with Frobenius action.** $H_{\text{ét}}^i(X_{\bar{\mathbb{F}}_q}, \mathbb{Q}_\ell)$ is a finite-dimensional vector space with a canonical Frob_q -action. For $\text{Spec } \mathbb{Z}$: no canonical cohomology theory produces a finite-dimensional vector space with a single global Frobenius action. (Arakelov cohomology and $\mathbb{F}_1$ -geometry are speculative replacements.)
2. **Hard Lefschetz.** Hard Lefschetz for $X/\mathbb{F}_q$ is geometric: the hyperplane class lives in $H_{\text{ét}}^2$ and the cup product is algebraic. For $\zeta(s)$: there is no algebraic variety, no hyperplane class, no cup product. The functional equation $s \leftrightarrow 1 - s$ plays the role of Hard Lefschetz, but it does not give the Hodge-type decomposition that controls eigenvalue sizes.
3. **Monodromy and Lefschetz pencils.** The propagation of purity through a family $X_t \rightarrow \mathbb{P}^1$ uses the monodromy representation and the geometry of singular fibers over $\bar{\mathbb{F}}_q$. For $\zeta(s)$: there is no family, no base curve, no singular fibers.

7 Summary: What Frobenius Gives and What Is Missing

Feature	Over $\mathbb{F}_q$	Over $\mathbb{Q}$
Frobenius	Global, canonical, geometric	Local only, not canonical
Galois group	$\hat{\mathbb{Z}}$ (abelian, simple)	$\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ (enormous)
Cohomology	Étale, finite-dim., Frob-equivariant	Conjectural (motives)
Purity	Proved (Deligne 1974)	Equivalent to RH
Hard Lefschetz	Proved	No algebraic analogue
Hilbert-Pólya H	$= \text{Frob}_q$ on H^1	Unknown

Theorem 12 (Weil as proof of concept). *The Weil conjectures prove RH for all smooth projective varieties over $\mathbb{F}_q$. The proof is a complete template: étale cohomology provides the Hilbert space, Frobenius provides the self-adjoint operator (via purity, which is the analogue of self-adjointness), and the Lefschetz trace formula connects the spectrum to the zeros of $Z(X, T)$.*

For $\zeta(s)$: the template fails at the first step (no canonical cohomology with a global Frobenius). RH for $\zeta(s)$ requires either:

- 1. A new cohomology theory for $\text{Spec } \mathbb{Z}$ admitting a global Frobenius-like operator (the motivic/absolute cohomology approach); or*
- 2. A direct construction of the Hilbert-Pólya operator (the spectral approach).*

These are equivalent: (1) would produce (2) by restricting to H^1 -cohomology; (2) would produce an operator serving as the “Frobenius” for $\zeta(s)$. Both are open.

References

- [1] Weil, A. (1949). Numbers of solutions of equations in finite fields. *Bull. Amer. Math. Soc.* 55, 497–508.
- [2] Dwork, B. (1960). On the rationality of the zeta function of an algebraic variety. *Amer. J. Math.* 82, 631–648.
- [3] Grothendieck, A. et al. (1972–1973). *SGA 4 et 4½*. Lecture Notes in Mathematics. Springer.
- [4] Deligne, P. (1974). La conjecture de Weil I. *Publ. Math. IHES* 43, 273–307.
- [5] Deligne, P. (1980). La conjecture de Weil II. *Publ. Math. IHES* 52, 137–252.
- [6] Montgomery, H. (1973). The pair correlation of zeros of the zeta function. *Proc. Symp. Pure Math.* 24, 181–193.

- [7] Odlyzko, A. (1987). On the distribution of spacings between zeros of the zeta function. *Math. Comp.* 48, 273–308.
- [8] Berry, M., Keating, J. (1999). The Riemann zeros and eigenvalue asymptotics. *SIAM Review* 41, 236–266.
- [9] Milne, J.S. (1975). On a conjecture of Artin and Tate. *Ann. Math.* 102, 517–533.